

An Auditable Agentic LLM Pipeline for Rubric-Based Grading: Evidence from a Real E-commerce Course Dataset

Nguyen Dinh Truc (corresponding) — 23521668@gm.uit.edu.vn

Nguyen Uyen Nhi — 23521115@gm.uit.edu.vn

Faculty of Information Systems, University of Information Technology (UIT), VNU-HCM, Vietnam

Supervisor: Trinh Trong Tin

Abstract

Rubric-based grading is essential for transparency and consistency but becomes a bottleneck in large classes. We present an auditable agentic grading pipeline implemented with the OpenClaw framework and apply it to four rubric-based assignments from an undergraduate e-commerce introduction course at UIT. The pipeline includes dataset normalization, text extraction with OCR when needed, incremental result writing, and automated auditing to guarantee one-to-one mapping between submissions and exported scores. Two human graders independently score a stratified subset of submissions; ground truth is defined as the mean of the two scores. We compare a single-agent grader (Phase 1) and a multi-agent grader (Phase 2) that decomposes grading into content/structure/language specialists and aggregates via a chairman agent. We report inter-rater agreement (QWK) and system-vs-human metrics (QWK/MAE/Pearson) with bootstrap confidence intervals, and analyze systematic score shifts and stability signals (veto/spread) induced by Phase 2. Results show that multi-agent grading does not automatically improve reliability and can introduce assignment-dependent calibration shifts; we provide practical recommendations for calibrating and aggregating multi-agent rubric grading in real classroom deployments.

Keywords: automated grading; rubric-based assessment; agentic AI; large language models; reliability; quadratic weighted kappa

1 Introduction

1.1 Motivation

Rubric-based assessment supports fairness and consistent grading, yet grading at scale remains time-consuming in large classes. Real classroom submissions are heterogeneous (DOCX/PDF/TXT and scanned artifacts), and instructors require auditable exports (no missing/duplicate records) for course administration and research use.

1.2 Gap

Automated scoring traditionally evaluates agreement against human raters using weighted kappa/QWK [1,2] and emphasizes validity and operational constraints [3]. Recent studies explore LLM-based rubric-grounded scoring [10], but classroom deployments require robust pipelines for heterogeneous files and OCR, auditability, and reliability-first evaluation. Multi-agent decomposition can help in some contexts but may also introduce calibration mismatch and aggregation instability [14,15].

1.3 Contributions

C1. An auditable grading pipeline (normalization → extraction/OCR → grading → incremental export → auditing) built on OpenClaw [18].

C2. We release derived, anonymized scoring artifacts (scores/metadata/audit summaries) and evaluation scripts via our public repository to enable independent verification of reported metrics.

C3. Reliability-first evaluation with human inter-rater QWK and AI-vs-GT (QWK/MAE/Pearson) with bootstrap confidence intervals.

C4. Evidence-based analysis of multi-agent grading: systematic score shifts, veto/spread stability signals, and practical calibration recommendations.

2 Related Work

2.1 Automated scoring and agreement metrics

AES/SAS research commonly measures system–human agreement using Cohen’s weighted kappa variants for ordinal scales [1] and practical interpretation guidelines [2]. Evaluation frameworks emphasize reporting reliability, validity, and operational constraints [3].

2.2 Benchmarks and modeling approaches

Benchmarks such as ASAP popularized QWK-based evaluation for essay scoring [7]. Neural AES methods are a strong baseline paradigm [8], and survey work summarizes modeling choices and evaluation practices [5,9].

2.3 LLM-based rubric-grounded grading

Prompt-based LLM scoring can be adapted to new rubrics without task-specific training, yet agreement varies by prompting and calibration. Recent work documents both promise and limitations in reliability and bias [10,11,12].

2.4 Multi-agent grading and aggregation

Multi-agent designs decompose grading into specialists and aggregate outputs. Such aggregation can introduce severity mismatch and instability without calibration [14]. Structured multi-agent scoring pipelines have also been proposed [15].

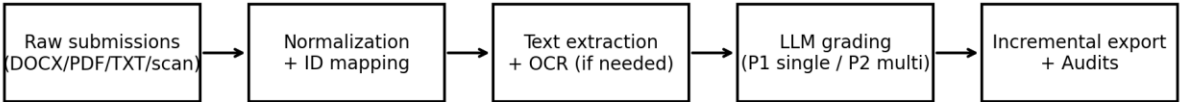
2.5 Reliability-first positioning

Automated scoring should be interpreted relative to human inter-rater reliability rather than as absolute truth [3]. Accordingly, we report human-human agreement as an upper bound and evaluate both phases against the human-mean reference, while quantifying uncertainty via bootstrap confidence intervals. This practice helps prevent over-claiming improvements when evaluation subsets are constrained and agreement estimates are noisy [2,16].

3 Method

Figure 1 shows the end-to-end pipeline.

Figure 1: OpenClaw grading pipeline (normalization → extraction/OCR → grading → export/audit).



OpenClaw Gateway (gpt-5.2 via Codex OAuth) enforces JSON-only outputs; rounding rules applied post-processing.

3.1 Implementation and model configuration

All LLM calls are served through a local OpenClaw Gateway exposing an OpenAI-compatible Chat Completions endpoint [18]. The gateway-reported model identifier in our deployment is `openai-codex/gpt-5.2` (a routing identifier); the underlying model weights are proprietary and not publicly disclosed. Therefore, we focus reproducibility on fixed decoding settings, JSON-only scoring schemas, released derived artifacts, and evaluation scripts rather than exact replication of model weights. Phase 1 uses temperature=0.1 and max_tokens=900; Phase 2 uses temperature=0.1 and max_tokens=800 with retry logic for rate limits and JSON parsing. Scores are post-processed with deterministic rounding (0.25 per criterion; 0.5 total; range 0–10).

4 Experimental Setup

4.1 Assignments and rubrics

Table 0 summarizes the four assignments and dataset sizes (full dataset).

Assignment	Topic (summary)	Typical format	N_full
------------	-----------------	----------------	--------

BT1	Study e-commerce curriculum & learning outcomes; factual completeness + sourcing	DOCX/PDF/TXT	137
BT2	Survey job market for 2 positions; analyze JD/skills/gaps	DOCX/PDF/TXT	129
BT3	Write 2 CVs for 2 positions; mapping CV↔JD + format/language	PDF/DOCX/images (OCR for scans)	123
BT4	Essay: personal career roadmap with required structure	PDF/DOCX	131

4.2 Human subset and metrics

Two human graders score a stratified subset (N=30 per assignment) selected to cover performance bands (high/medium/low) based on Phase-1 scores. Ground truth (GT) is defined as the mean of the two human scores. We report inter-rater QWK and AI-vs-GT metrics (QWK/MAE/Pearson) with 95% bootstrap CIs.

5 Results

5.1 Human agreement + Phase 1 (Table 1a)

Table 1a reports human agreement and Phase 1 vs GT with 95% bootstrap confidence intervals (N=30 per assignment).

BT	Human QWK [CI]	P1 QWK [CI]	P1 MAE [CI]	P1 r [CI]
BT1	0.698 [0.464, 0.811]	0.484 [0.273, 0.627]	1.108 [0.892, 1.333]	0.808 [0.681, 0.892]
BT2	0.684 [0.469, 0.802]	0.331 [0.164, 0.443]	1.375 [1.142, 1.600]	0.694 [0.411, 0.860]

BT3	0.671 [0.440, 0.835]	0.827 [0.664, 0.904]	0.775 [0.550, 1.042]	0.897 [0.807, 0.960]
BT4	0.628 [0.301, 0.791]	0.720 [0.587, 0.799]	0.575 [0.442, 0.725]	0.753 [0.630, 0.861]

5.2 Phase 2 (Table 1b)

Table 1b reports Phase 2 vs GT with 95% bootstrap confidence intervals (N=30 per assignment).

BT	P2 QWK [CI]	P2 MAE [CI]	P2 r [CI]
BT1	0.043 [-0.026, 0.142]	1.842 [1.517, 2.158]	0.225 [-0.115, 0.557]
BT2	-0.002 [-0.237, 0.244]	0.958 [0.708, 1.250]	-0.010 [-0.356, 0.337]
BT3	0.613 [0.405, 0.747]	1.058 [0.833, 1.300]	0.673 [0.509, 0.837]
BT4	0.517 [0.262, 0.698]	0.608 [0.467, 0.750]	0.553 [0.324, 0.725]

5.3 Score shifts and stability signals (Table 2)

BT	N_full	mean(P2-P1)	median(P2-P1)	veto rate	mean spread
BT1	137	-0.836	-1.000	0.993	5.150
BT2	129	1.787	2.000	0.922	3.387
BT3	123	-0.943	-1.000	0.772	2.653
BT4	131	-0.889	-1.000	0.061	1.133

5.4 Visualization (Figures 2a/2b)

Figure 2a: Phase 1 vs GT scatter plots (N=30 per assignment).

Figure 2a. Phase 1 (single-agent) vs GT (N=30 per assignment)

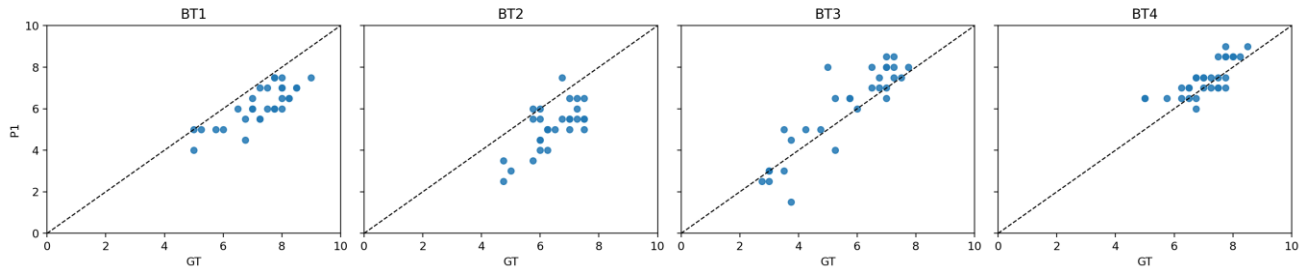
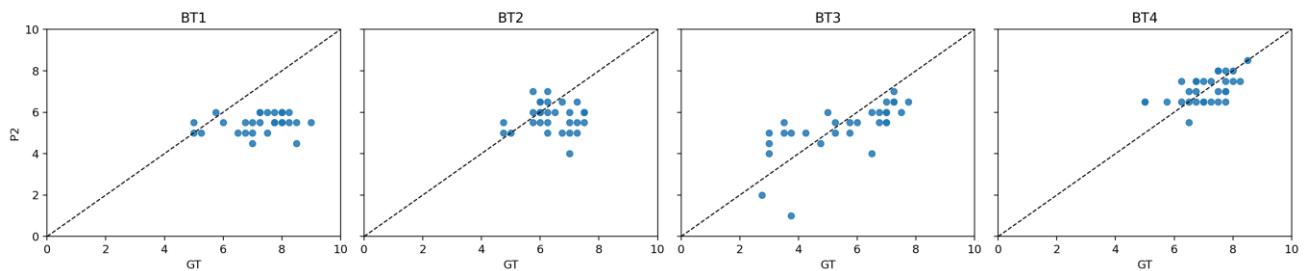


Figure 2b: Phase 2 vs GT scatter plots (N=30 per assignment).

Figure 2b. Phase 2 (multi-agent) vs GT (N=30 per assignment)



References

- [1] Cohen, J.: Weighted kappa: Nominal scale agreement with provision for scaled disagreement or partial credit. *Psychological Bulletin* 70(4), 213–220 (1968).
- [2] Landis, J.R., Koch, G.G.: The measurement of observer agreement for categorical data. *Biometrics* 33(1), 159–174 (1977).
- [3] Williamson, D.M., Xi, X., Breyer, F.J.: A framework for evaluation and use of automated scoring. *Educational Measurement: Issues and Practice* 31(1), 2–13 (2012).
- [4] Shermis, M.D., Burstein, J. (eds.): *Handbook of Automated Essay Evaluation: Current Applications and New Directions*. Routledge (2013).
- [5] Beigman Klebanov, B., Madnani, N.: *Automated Essay Scoring*. Synthesis Lectures in Human Language Technologies. Morgan & Claypool (2021).
- [6] Attali, Y., Burstein, J.: Automated essay scoring with e-rater® V.2. *Journal of Technology, Learning, and Assessment* 4(3) (2006).
- [7] Hewlett Foundation (ASAP). Automated Student Assessment Prize (ASAP) AES benchmark (Kaggle competition). <https://www.kaggle.com/c/asap-aes> (accessed 2026-05-01).
- [8] Taghipour, K., Ng, H.T.: A neural approach to automated essay scoring. In: *Proceedings of EMNLP 2016*, pp. 1882–1891. ACL (2016).

- [9] Ke, Z., Ng, V.: Automated essay scoring: A survey of the state of the art. In: Proceedings of IJCAI 2019, pp. 6300–6308 (2019).
- [10] Mizumoto, A., Eguchi, M.: Exploring the potential of using an AI language model for automated essay scoring. *Research Methods in Applied Linguistics* 2(2), 100050 (2023). <https://doi.org/10.1016/j.rmal.2023.100050>
- [11] Esterhuysen, H., et al.: Large language models in student assessment: Comparing ChatGPT and human graders. *arXiv:2406.16510* (2024).
- [12] Meyer, L., et al.: Reliable but not rigorous: Evaluating ChatGPT's reliability, validity, and bias in automated academic grading. *Computers and Education: Artificial Intelligence* (2025).
- [13] Hakala, K., et al.: Automated refinement of essay scoring rubrics for language models via Reflect-and-Revise. *arXiv:2510.09030* (2025).
- [14] Seßler, K., et al.: Specialists or generalists? Multi-agent and single-agent LLMs for essay grading. *arXiv:2601.22386* (2026).
- [15] Wang, Y., et al.: AutoSCORE: Enhancing automated scoring with multi-agent large language models via structured component recognition. *arXiv:2509.21910* (2025).
- [16] Fleiss, J.L., Cohen, J.: The equivalence of weighted kappa and the intraclass correlation coefficient as measures of reliability. *Educational and Psychological Measurement* 33(3), 613–619 (1973).
- [17] OpenAI: API reference for Chat Completions (accessed via internal proxy in this work). <https://platform.openai.com/docs/api-reference/chat> (accessed 2026-05-01).
- [18] OpenClaw Framework. Documentation: <https://docs.openclaw.ai> ; Source: <https://github.com/openclaw/openclaw> (accessed 2026-05-01).